

Supplementary Material

Software and artificial intelligence for reference evapotranspiration in agriculture: A
systematic review

This Supplementary Material collects the descriptive bibliometric figures and tables that support, but are secondary to, the taxonomic and paradigmatic synthesis in the main text. Figures are generated by the reproducible Python pipeline; the underlying tables are provided as CSV files in the project repository (**results/tables/**). **All values will be regenerated on the final included set once title/abstract screening is complete.**

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S1 Scientific production and keywords

Author-keyword frequency (Fig. [S1](#)), a keyword word cloud (Fig. [S2](#)) and a keyword tree map (Fig. [S3](#)) summarise the thematic composition of the corpus.

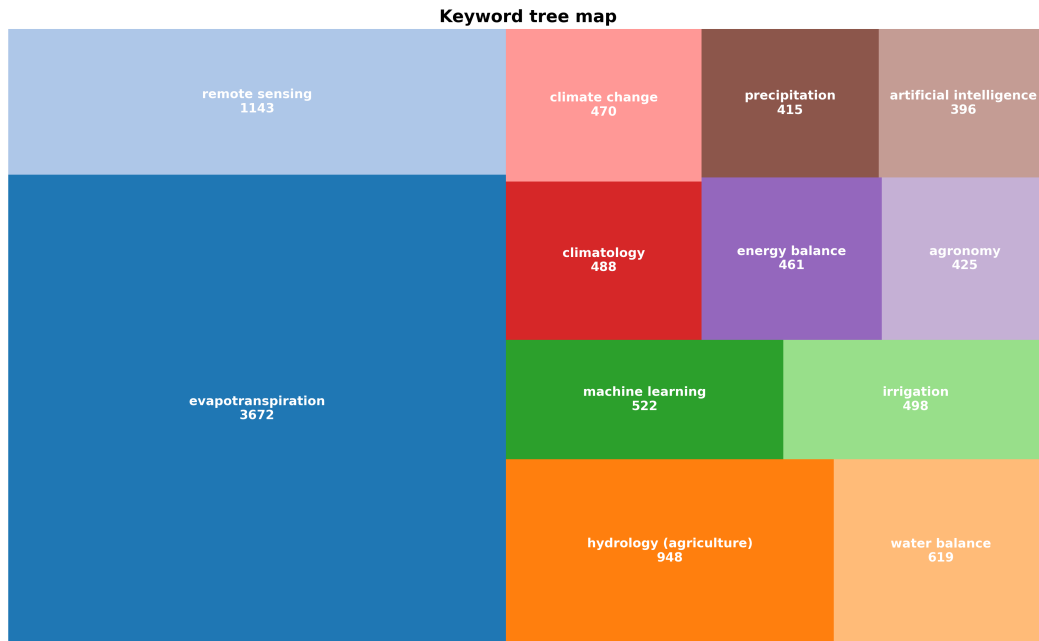


Figure S3: Tree map of author-keyword groups.

S2 Geography and international collaboration

Leading countries by output (Fig. S4) and choropleth maps of documents (Fig. S5) and citations (Fig. S6) per country describe the geographical distribution of research; the international co-authorship map (Fig. S7) shows collaboration links between countries.

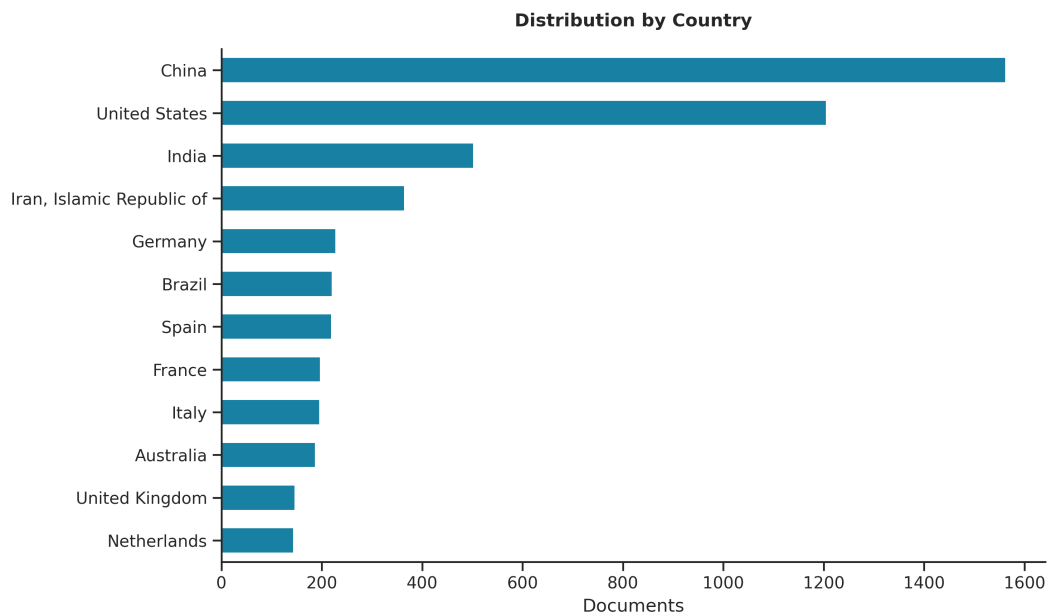


Figure S4: Leading countries by number of documents (author affiliations).

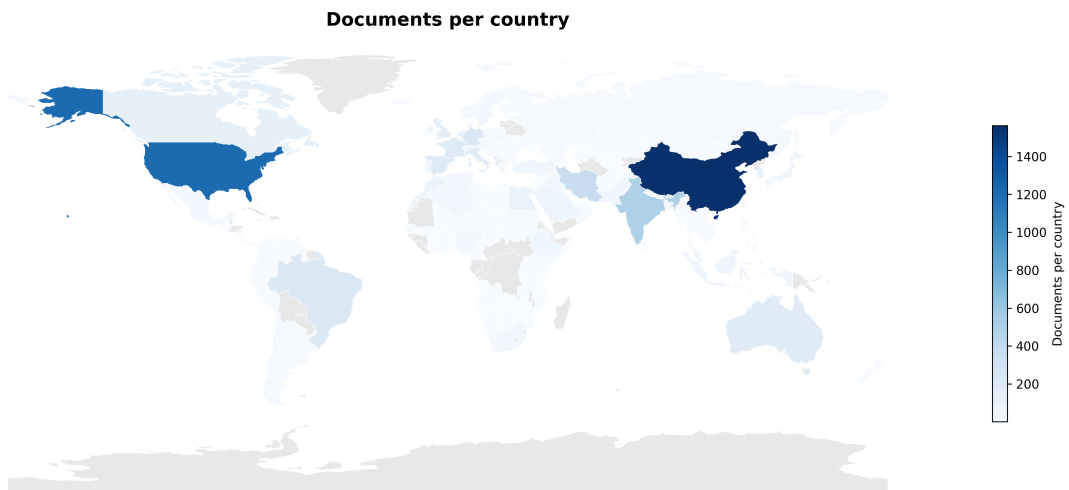


Figure S5: Choropleth map of documents per country.

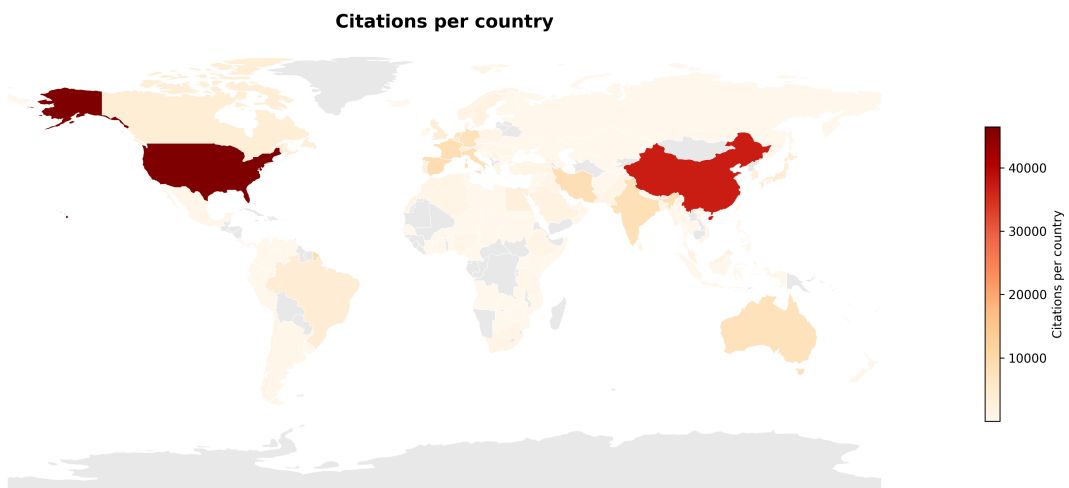


Figure S6: Choropleth map of citations per country.

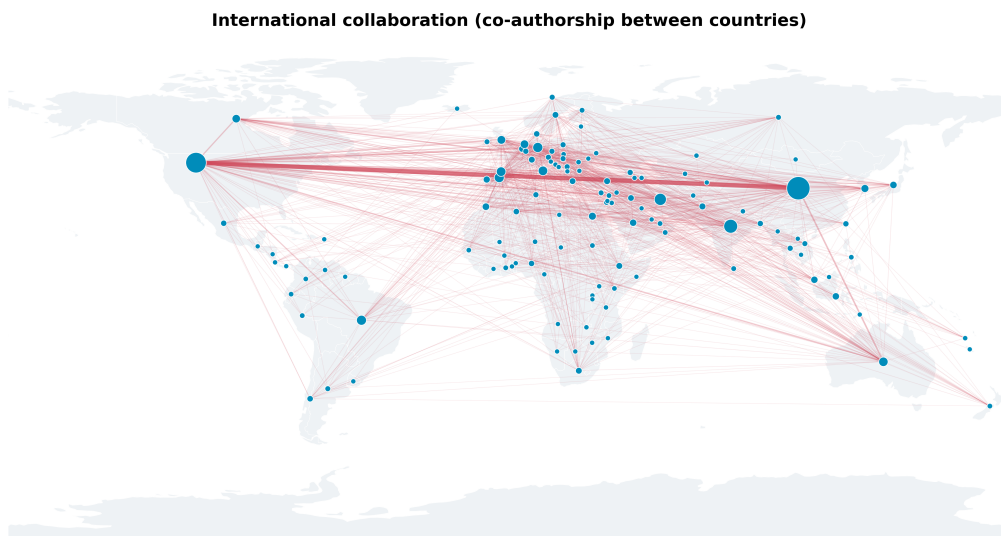


Figure S7: International co-authorship collaboration map.

S3 Author and keyword networks; thematic structure

The author co-authorship network (Fig. S8), the strategic (thematic) map of keyword clusters (Fig. S9) and the thematic evolution across sub-periods (Fig. S10) characterise the intellectual structure of the field. Interactive versions of the network and thematic-evolution figures are available in the repository as HTML ([fig8_author_network.html](#), [fig11_thematic_evolution.html](#)).

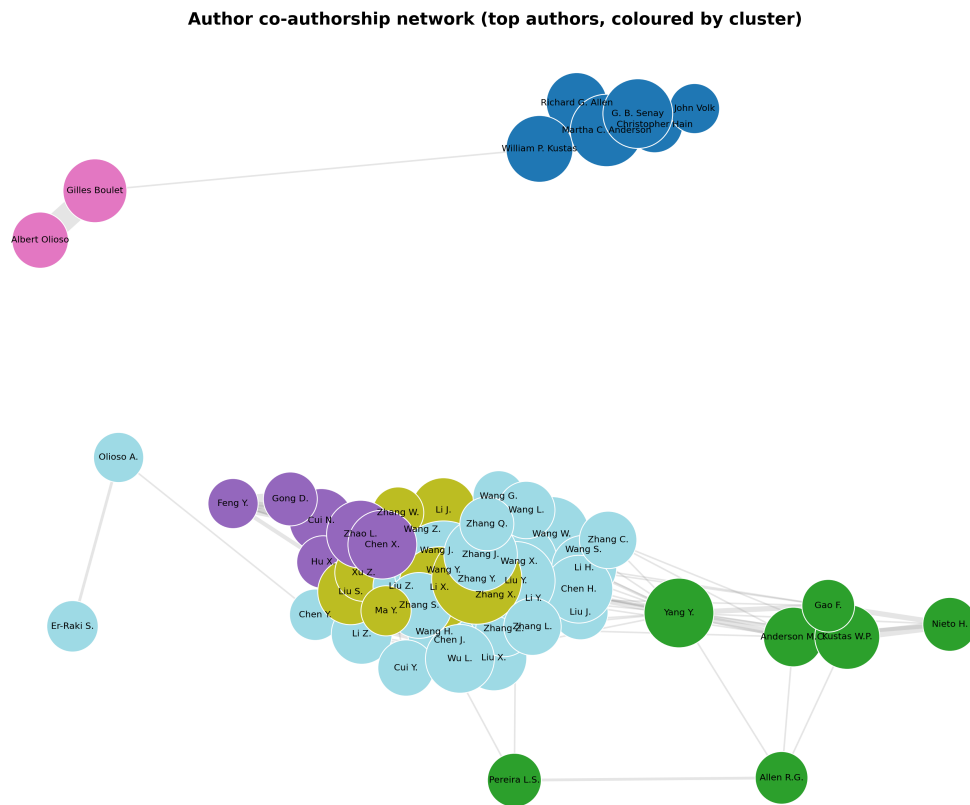


Figure S8: Author co-authorship network with community detection.

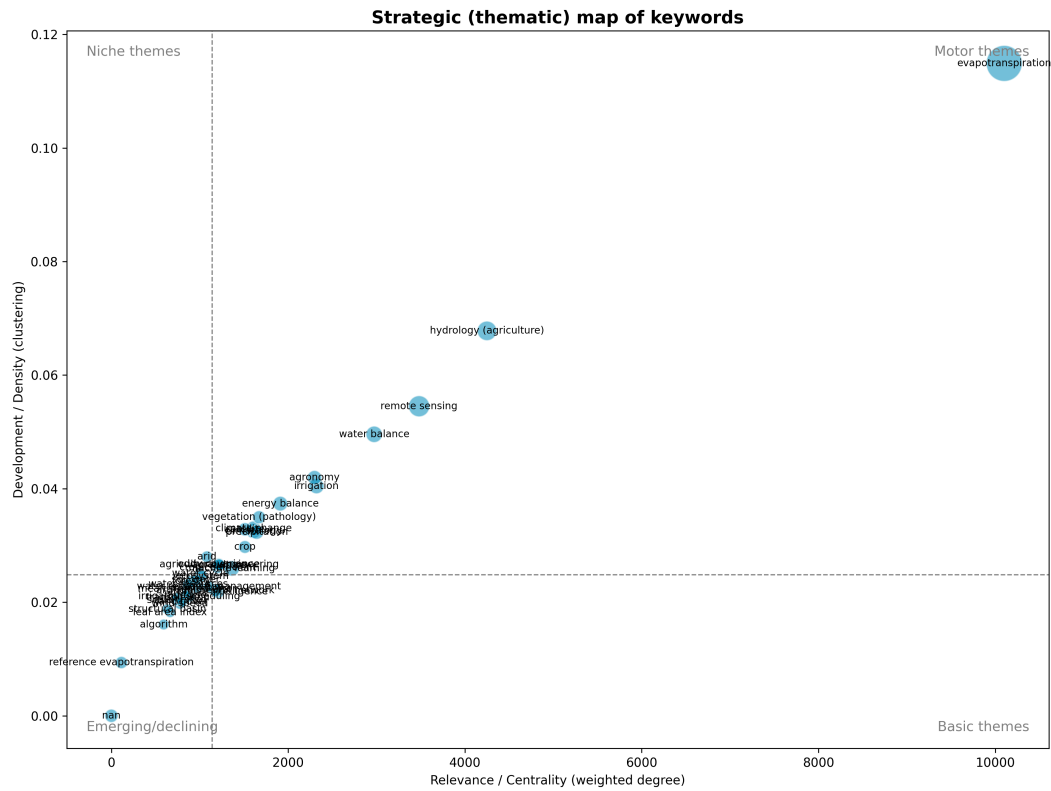


Figure S9: Strategic (thematic) map of keyword clusters.

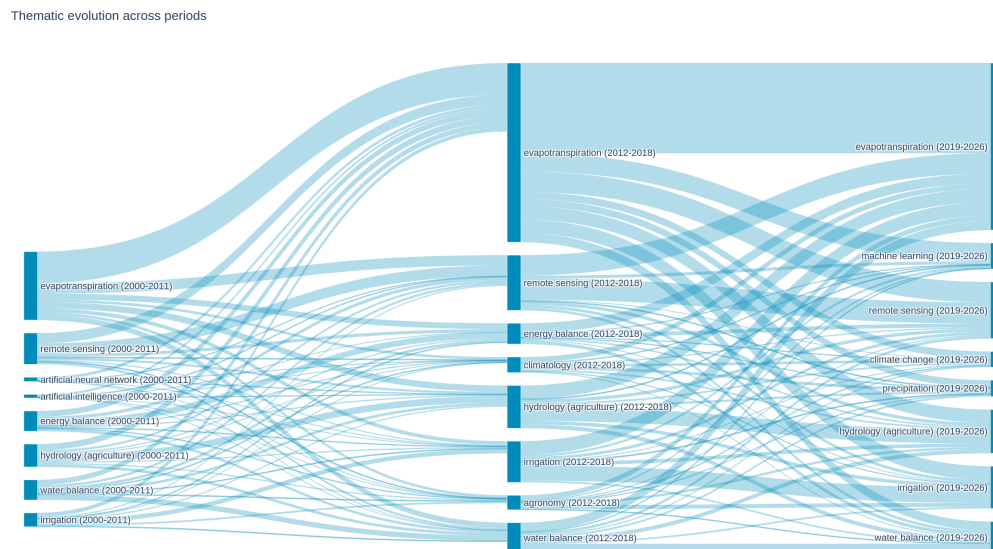


Figure S10: Thematic evolution (Sankey) across sub-periods.

S4 Institutions, journals and open access

The most productive institutions (Fig. S11), a journal output-versus-impact chart (Fig. S12) and the open-access breakdown (Fig. S13) situate the corpus in terms of contributing organisations, publication venues and accessibility.

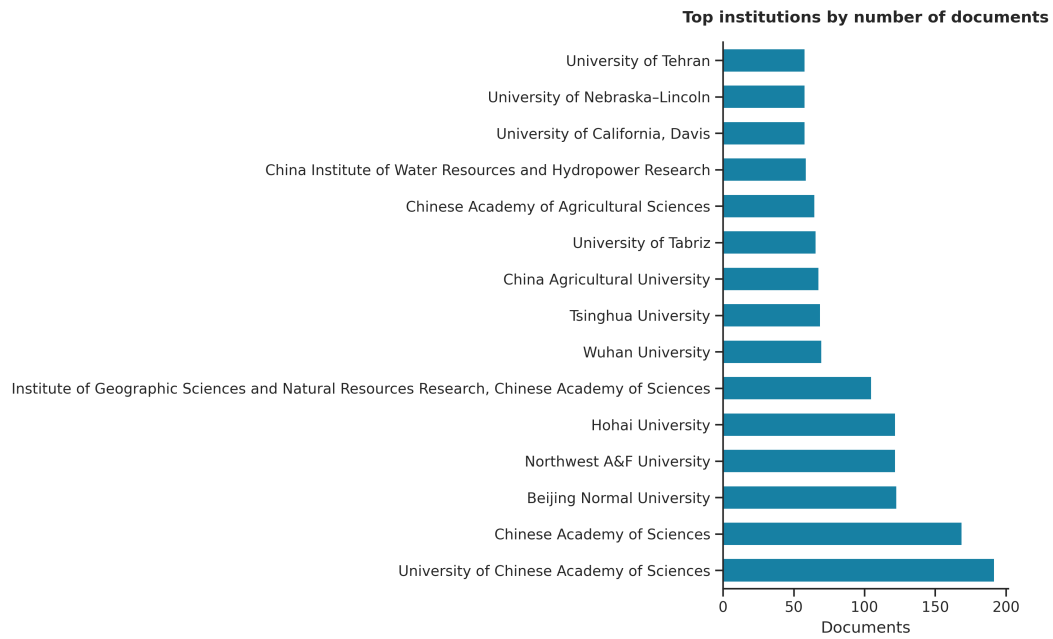


Figure S11: Most productive institutions (author affiliations).

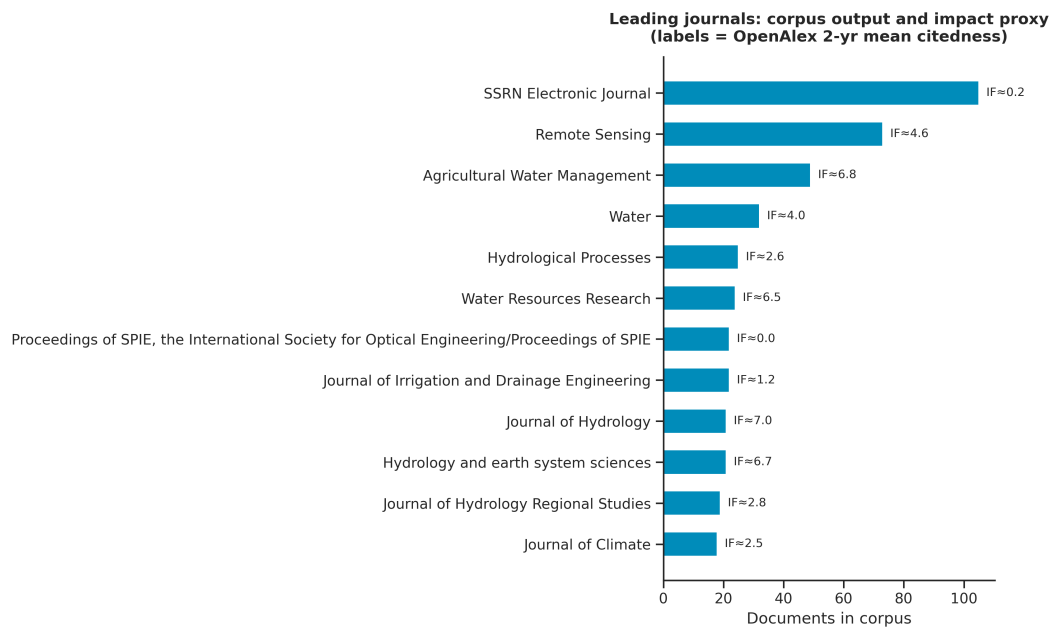


Figure S12: Journal output versus impact (OpenAlex 2-year mean citedness).

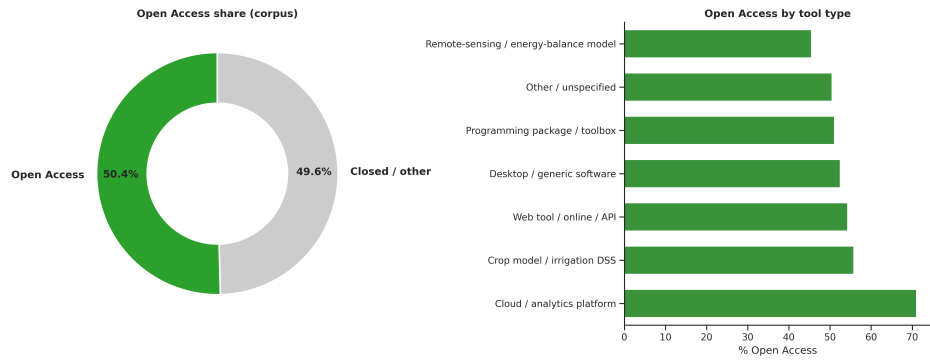


Figure S13: Open-access status of the corpus.

S5 Subject fields and deterministic/probabilistic contrast

The distribution of subject categories (Fig. S14) and a macro-level contrast between deterministic and probabilistic irrigation/ET₀ work (Fig. S15) complete the descriptive picture.

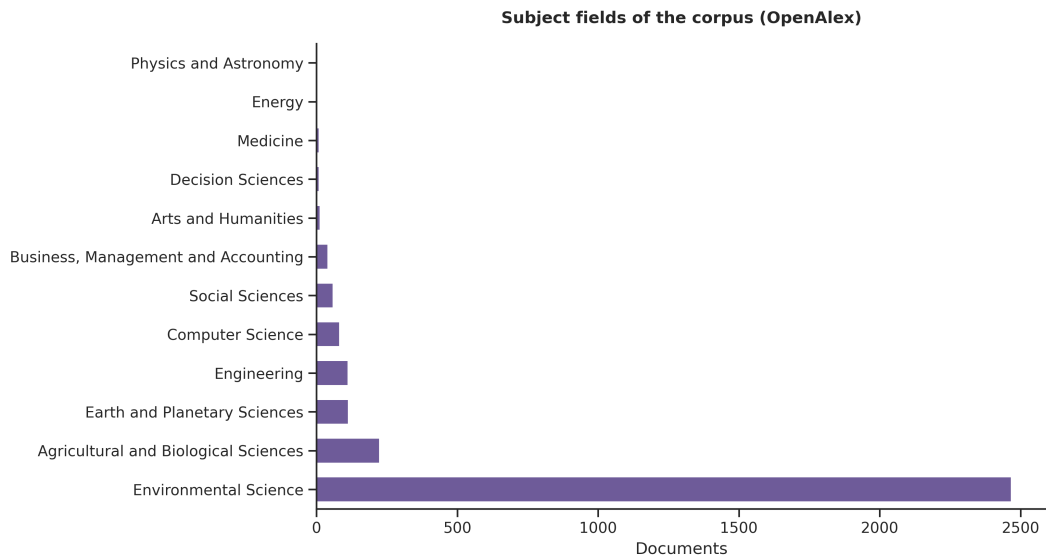


Figure S14: Distribution of documents by subject category (OpenAlex topics).

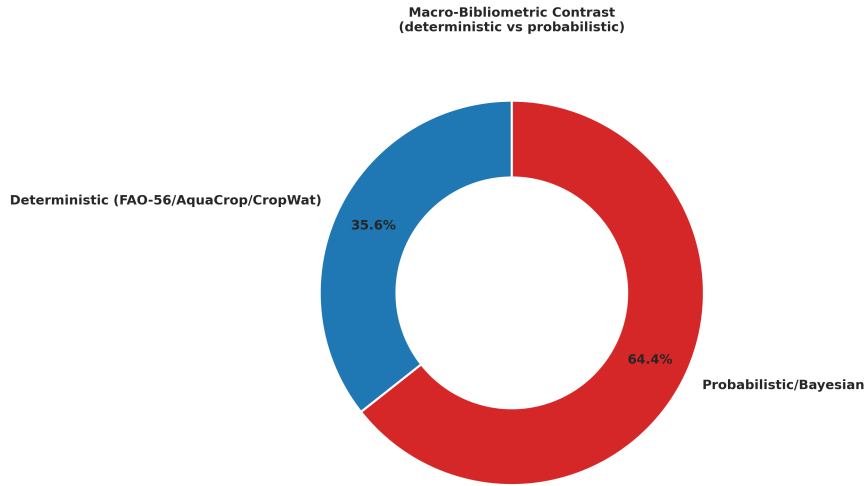


Figure S15: Macro-level contrast of deterministic versus probabilistic irrigation/ET₀ documents.

S6 Supplementary tables (provided as CSV)

The following tables are provided in full as CSV files in **results/tables/** and will be finalised on the included set:

Table S1: Index of supplementary tables available in the repository.

File	Content
publications_per_year.csv	Annual scientific production
top_countries.csv	Leading countries by output
top_institutions.csv	Most productive institutions
top_journals.csv	Leading journals by output
top_journals_by_impact.csv	Journals ranked by mean citedness
journal_metrics.csv	Per-journal output and impact proxy
top_keywords.csv	Author-keyword frequencies
tool_type_distribution.csv	Tool-type taxonomy counts
method_classification.csv	Computational-paradigm counts
method_by_year.csv	Paradigm shares over time
ai_technique_distribution.csv	AI-technique breakdown
subject_categories.csv	Subject-category counts
oa_summary.csv	Open-access summary
oa_by_tool_type.csv	Open access by tool type
gap_oceanic_indices.csv	Teleconnection/prior mentions
top_cited_overall.csv	Most-cited documents
source_overlap.csv	Database provenance/overlap